

SETHULAKSHMI BABU

Software Engineer Intern | Data & AI Specialist

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PROFESSIONAL PROFILE

Analytical M2 Engineering student at **ESILV Graduate School** specializing in Data and AI. Expert in building scalable backend tools and **Agentic AI** workflows using **Python** and **Java**. Experienced in real-time ML pipelines, automated data preparation, and **MLOps**. Proactive problem-solver dedicated to delivering qualitative, well-documented code for enterprise-grade AI ecosystems.

TECHNICAL EXPERTISE

Programming & Scripting: Python (Pandas, Scikit-learn), Java (Maven, IntelliJ), SQL (PostgreSQL, MySQL, Oracle), C++, JavaScript, TypeScript, Node.js, Bash/Shell scripting.

Machine Learning & AI: Random Forest, XGBoost, Logistic Regression, CNNs, Feature Engineering, Model Evaluation, Outlier Detection, Prompt Engineering (Vertex AI), Chain-of-Thought.

Deep Learning & LLMs: TensorFlow, Keras, Vertex AI Pipelines, LLM Fine-tuning, LangChain, Ollama, Agentic AI, BERT Embeddings, Semantic Search, Multi-step Task Orchestration.

Infrastructure & Tools: Docker, Kubernetes, CI/CD (GitHub Actions), Git, Pydantic (Data Validation), ETL Pipelines, EDA, Anomaly Detection, Time-series Analysis.

Frontend & Visualization: React, HTML, CSS, Streamlit, Plotly, Matplotlib, Excel, MS Office Suite.

SOFTWARE ENGINEERING & AI PROJECTS

Bike-PIM Agentic Scraper | Unstructured Data & Files Processor

[GitHub](#)

- Developed an agentic Python pipeline to interpret and extract unstructured web product data into validated JSON schemas.
- Leveraged LLMs for context-aware parsing and **Pydantic** for rigorous schema integrity and enterprise-grade ingestion.
- Optimized scraping resilience with anti-bot evasion and simulation fallbacks for complex e-commerce environments.

Path Optimization AI | Geospatial Routing & Geo-Grouping

[GitHub](#)

- Engineered a routing optimizer utilizing enhanced **A-star algorithms** and dynamic penalty logic to minimize travel costs.
- Integrated real-time APIs for adaptive route generation applicable to logistics networks and **geo-grouping** use cases.

Intelligent Document Auditor | NLP & Tamper Detection

[GitHub](#)

- Designed an inspection pipeline using LLMs and embeddings to detect semantic inconsistencies in unstructured repositories.
- Implemented RAG-based retrieval for context-aware anomaly scoring, providing explainable flags for automated auditing.

Smart AI Resume Matcher | Unsupervised Modeling & Search

[GitHub](#)

- Built an unsupervised engine using **BERT Embeddings** to automate the alignment of resumes with job descriptions.
- Developed weighted scoring algorithms for semantic similarity, supporting advanced search and filtering operations.

Medipatient AI | Clinical ML Pipeline & Decision Support

[GitHub](#)

- Integrated Random Forest classification with LLM reasoning to interpret patient data and provide explainable diagnostics.
- Developed end-to-end Python pipelines with leakage prevention, ensuring **ethical AI practices** for real-world healthcare data.

Drone Health Guard | Predictive Maintenance & Anomaly Detection

[GitHub](#)

- Engineered an end-to-end ML pipeline for real-time sensor-level fault detection from multivariate telemetry data.
- Participated in a technical Hackathon hosted by **ESILV and IBM**, focusing on model precision and low-latency inference.

AI-Powered Smart Analytics Dashboard | Full-Stack Data App

[GitHub](#)

- Built an interactive analytics platform with **Streamlit** and **Plotly** for dynamic visualization of heterogeneous datasets.
- Developed modular ETL pipelines for rapid ingestion, converting raw data into actionable business intelligence.

EDUCATION

Master of Science in Engineering Major: Data and AI

2024 – Present

ESILV Engineering School, Courbevoie, France | Focus: Production-ready LLM engineering, Python.

Bachelor of Science in Computer Science

Graduated 2023

Mahatma Gandhi University, India | Grade: 88.80% (Distinction)

CERTIFICATIONS & COLLABORATION

- Certifications:** watsonx.data Technical Essentials (2025); Ethical Hacking (2023); Agile Methodology (2024).
- Languages:** English (C1 Proficiency), French (B1 Proficiency), Malayalam (Native).
- Soft Skills:** Analytical, detail-oriented, collaborative problem-solver; autonomous project management.